Access and Edge Facility-Class Design Basis

The Second Facility Class of the Communications Sector Design Basis

DBA-COMM-AC-FC2 — Facility-Class Design Basis, DBA-COMM Series

Issued Under: Communications Sector Framework (DBA-COMM_Sector_Framework v0.14); DBA-MA Parent Framework

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose and Scope

1.1 Purpose

This document sets out the design basis for the access and edge of the communications network: the cell sites, central offices, local lines, user terminals and broadcast transmitters through which service is finally handed to people and to the facilities that depend on it. It is the second facility class in the DBA-COMM series. The *Communications Sector Framework* (DBA-COMM_Sector_Framework) states the sector's design basis in general terms, and this document applies it to access. It names the events an access facility must withstand, traces how each reaches the people and sectors that depend on it, sets the measures that size its protection, and states who must authorize what. It describes no particular carrier or site. A design basis for a specific site or service area starts from this document and adds only what is particular to it.

Access is where the network meets the people who use it. A 911 call, an emergency alert, a hospital's connection and a grid operator's link to a substation all leave the network through an access facility, and very often through only one. Access is also the part of the network that runs out of power first when the grid fails. Those two facts, together, are what this class is about.

1.2 Scope

The class covers the facilities that deliver service in a local area: cell sites, meaning the towers, rooftop antennas and small cells with their base stations, batteries, generators and backhaul links; wireline central offices and the remote cabinets they feed, with the copper, fiber and coaxial lines to homes and businesses; fixed-wireless and satellite user terminals and the satellite networks that serve them; and over-the-air broadcast transmitters for television and radio, including the stations that carry the Emergency Alert System. It covers the systems that monitor and manage those facilities remotely, and the people who operate, fuel and repair them. On the author's decision of 7 October 2026 the user terminals in scope include mobile handsets, for their emergency-call and alert-receiving functions only; the rest of a handset's design is outside the class. The class also carries the dissemination of public alerts: over broadcast stations through the Emergency Alert System, and over carriers' cell networks through Wireless Emergency Alerts and cell broadcast, on the author's decision of 6 October 2026 that origination is the emergency services coordination class's and dissemination is this class's (Emergency Management and Coordination §8.3). FirstNet, the carrier network built for public safety, is in the class with the elevation of Section 8.1, and counts toward a critical dependent's second path only where Section 8.1's test is met (*Nashville Bombing* F-7).

The class stops at three boundaries. On the author's decision of 6 October 2026, a central office building and its backup power belong to this class, and the switching and routing decisions made inside it belong to the Core class, DBA-COMM-CR-FC3 (*Core Switching, Routing and Signaling*). Paths that carry traffic between a central office or cell site and the backbone belong to the Transport class, DBA-COMM-TR-FC1 (*Transport and Backbone*), from the point where

local traffic is gathered onto them. The reception and distribution of timing, on which cell sites depend, belongs to DBA-COMM-PN-FC4. The boundaries are developed in Section 8.3. Two further boundaries are drawn by the emergency services series. The 911 system begins where the call and the caller's location enter it, so a 911 center and, in a Next-Generation 911 system, the Emergency Services IP Network and its core services are that series', and this class protects the access paths that reach them (Public-Safety Answering and Dispatch §8.3). And the origination of an alert is the alerting authority's, while its dissemination is this class's (Section 4.7).

1.3 The Reference Configurations

Three configurations are used as reference cases.

A cell site, with its batteries, any generator, its backhaul link and the area it covers, is the governing configuration. There are hundreds of thousands of them, each small, each dependent on grid power, and together they carry most of the calls people make in an emergency. It sets the class's first measure.

A wireline central office serving a 911 call center and a hospital shows the second configuration. It is one building, often decades old, through which many lines and many critical dependents pass, and it is the access facility whose loss reaches farthest. Nashville was such a building, and its reach ran beyond its own dependents, to the backhaul of other sites, another carrier's traffic, FirstNet and the State's network (*Nashville Bombing* F-4); Section 3.4 counts that reach.

A broadcast transmitter carrying emergency alerts shows the third. It is the path by which warnings reach people who have no working phone, and it depends on its own power and its own link to the alert originator. After Hurricane Maria it was the path that remained, and broadcasters did not report their own state (*Hurricane Maria* F-8).

A requirement that names one configuration applies to that configuration. A requirement that names none applies to all of them.

1.4 Class Type

This is a physical class under the author's determination of 30 September 2026 (*Facility-Class Types Are Physical, Coordination and Logical*). The harm it can do comes mainly from the loss of physical things: a site, a building, a line, a terminal, and the power that keeps them running. Its systems of remote management matter, and Section 4.6 treats them, but they matter because they can take many sites or terminals out of service at once.

1.5 Relationship to Governing Documents

This instrument is issued under the *Communications Sector Framework* and inherits the DBA-MA parent framework. It does not repeat the framework's taxonomy, principles or Bright Line and Command Broker architecture; it inherits them, and Section 9 is the required inheritance map. Where the framework is still a skeleton (framework §4, §5 and §7), this instrument develops the class-level content ahead of it and records in Section 10 what the framework should

take up. References of the form “framework §N” are to the *Communications Sector Framework* in force.

Governing instruments. This instrument does not own the concepts it applies. The Bright Line — its placement criterion, its granularity, its scope of application and the obligation it creates — is defined in FD-BL (*The Bright Line*), which governs. The Broker role that discharges that obligation is defined in FD-BR (*The Broker*). Layer decomposition, the determinism boundary as a scope gate, and the machine-learning carve-out are defined in FD-LD (*Layer Decomposition and Determinism*). Local sovereignty, the property of a protective function that executes independently of reachable digital authority, is defined in FD-LS (*Local Sovereignty*). Two determinations bind under FD-BL: FD-BL-D1 (*Mode of Discharge*) governs how the authorization obligation may be discharged, and FD-BL-D2 (*Indication Integrity*) governs the integrity of the basis on which an authorization is granted. FD-EV (*Envelope*) reserves the term “envelope”. Placement is on the consequence of the individual action, assessed per action; this instrument applies that definition and does not vary it. The Foundational Definitions are cited without a version token, and FD-REG (*Governing Stack Register*) is authoritative for their version state.

The communications design basis is cited, not inherited. DBA-MA-CDB (*Military Action Communications Design Basis*) sets out seven requirements, CDB-1 through CDB-7, for every facility in every sector whose communications may be degraded or denied. An access facility depends on communications for its own monitoring and management, and Section 7 states how the seven requirements apply to it, as *Military Action Communications Design Basis* §3 requires (framework §1.5, §8.1).

2. The Access and Edge Facility Class

2.1 Defining Characteristics

Three features define the class.

First, it depends on power more than any other part of the network. A central office typically carries batteries and a generator for eight to twenty-four hours, a figure that counts the generator; a cell site typically carries batteries for four to eight hours and often no generator (framework §3.3). This class states a central office’s figure as a battery figure and a generator figure separately, because a generator not protected against an event counts as none for that event (Section 3.2): against the loss of the office itself its endurance is its batteries’, about four to five hours at Nashville on the differing accounts of its record (*Nashville Bombing* F-3). When the grid fails over a wide area, every access site in that area begins to run down at the same moment, and the network’s coverage shrinks on a timetable set by the batteries.

Second, it is where the network’s outward criticality concentrates. A 911 call center, a hospital, a water plant’s remote stations or a substation’s control link are each served by a small number of access facilities, often one. The loss of that one is the loss of the service, whatever the state of the

network behind it. And one facility may carry more than its own dependents: the backhaul of other sites, other carriers' traffic and other providers' lines can pass through it (*Nashville Bombing* F-4).

Third, it is numerous and remotely managed. The class consists of very many small facilities and very many user terminals, managed from a few central systems. A single remote act from one of those systems can reach thousands of sites or tens of thousands of terminals at once. The reach of a single act is the management partition it can enter: at KA-SAT one set of commands reached the terminals of one consumer partition across a continent and none outside it (*KA-SAT Attack* F-1).

2.2 Deployment Typologies and Size Range

The class spans a wide range. At one end is a small cell on a lamppost covering a few city blocks. At the other is a central office serving tens of thousands of lines and several critical dependents, or a satellite network serving terminals across a continent. Between them are the macro cell sites that cover most of the country, the remote cabinets of the wireline network, and the broadcast transmitters that cover whole regions. The class design envelope applies to all of them, and the values of its measures differ widely across them.

2.3 Consequence Context

The class's governing harm is the loss of service across an area, and with it the loss of the ability to call for help and to be warned. Its second harm is the loss of a single access facility on which a critical dependent relies. Its third is the loss of many sites or terminals at once through a common system of management.

Three events are the reference events for the class: the Nashville bombing, Hurricane Maria and the KA-SAT attack. The class proposed them, and the framework confers them at its §9.1 as DBA-COMM-RE-4 to DBA-COMM-RE-6, under the author's decision of 6 October 2026 that every reference event a class proposes is conferred. Their instruments are written, and under the determination of 9 October 2026 that gives reference events their own project they are held by Project 11. This revision takes in what they return, and the accounts below are conformed to them (Section 10.5). They are cited by the short titles *Nashville Bombing*, *Hurricane Maria* and *KA-SAT Attack* and finding, recommendation or section number.

The Nashville bombing of 25 December 2020 (DBA-COMM-RE-4) is the concentrated case. A vehicle bomb exploded at about 06:30 that morning in the street in front of AT&T's central office on Second Avenue North in downtown Nashville. The building withstood the blast, and its equipment ran on batteries for about four to five hours after its commercial power connection was severed and its generators disabled. AT&T's personnel were kept from the building for about twelve hours. 911 failed from about 11 a.m., and service failed across parts of Tennessee, Kentucky and Alabama shortly after noon, with effects reported in at least seven states; the failure came when the batteries ran out, not at the explosion. Sixty-six Tennessee answering

points or districts lost 911, for up to 97.5 hours on the Tennessee Emergency Communications Board's count, and the last answering point was restored on 1 January 2021 on the Cybersecurity and Infrastructure Security Agency's account. The Federal Bureau of Investigation determined that the act was not related to terrorism. One building carried far more than its own neighborhood, and the building stood while its power plant did not.

Hurricane Maria in September 2017 (DBA-COMM-RE-5) is the power case. The Federal Communications Commission's Disaster Information Reporting System showed 95.2 per cent of the cell sites reported in Puerto Rico out of service the day after landfall and 95.6 per cent at the worst, on 23 September 2017, with 76.6 per cent in the Virgin Islands. Outages persisted for months, not weeks: the cell sites took six months to reach 4.3 per cent out, and wireline and cable remained generally out into December for lack of power, while the grid was rebuilt. On the Government Accountability Office's analysis, about half the outages in September were for want of power and the rest for damage or loss of network connection, and about four-fifths were for power by October. The storm's damage was as large as the loss of power at the event's onset; over its duration, power and backhaul did most of the harm.

The attack on Viasat's KA-SAT network on 24 February 2022, at the start of Russia's full-scale invasion of Ukraine (DBA-COMM-RE-6), is the management case. On Viasat's report of 30 March 2022, an attacker who had entered the management segment through a VPN appliance executed legitimate management commands that overwrote the flash memory of a large number of residential modems of one consumer partition. The modems left the network from about 04:15 UTC, after a flood of malicious traffic from modems in Ukraine that began at about 03:02 UTC and, on the operator's account of 2023, continued for weeks. Several thousand customers in Ukraine were affected, on Viasat's report, and tens of thousands across Europe, on press reports, among them nearly 9,000 of Nordnet's satellite subscribers on Orange's count and about a third of Bigblu Broadband's 40,000 subscribers in six countries on Eutelsat's confirmation as AFP reported it (*KA-SAT Attack* §3.4). About 5,800 Enercon wind turbines in central Europe lost remote monitoring and control and kept running in automatic mode, on Enercon's account, because they did not need the link to run, and that is part of the lesson; their monitoring was lost for weeks (Section 8.1).

The attribution of the KA-SAT attack is stated as each body's, in its own words, and this class makes none of its own. The European Union, in the High Representative's declaration of 10 May 2022, condemned "the malicious cyber activity conducted by the Russian Federation against Ukraine", which "targeted the satellite KA-SAT network, owned by Viasat". The United Kingdom's Foreign, Commonwealth and Development Office said that the National Cyber Security Centre "assesses that it is almost certain Russia was responsible". The United States' Secretary of State said that "Russia launched cyber attacks in late February against commercial satellite communications networks to disrupt Ukrainian command and control during the invasion". Canada's ministers said that "Canada assesses that the Russian military was behind this incident"; Australia's ministers that, "together with our partners, we assess that Russia

launched cyber attacks in late February against commercial satellite communications networks"; and Estonia joined "the coalition attributing the 24 February cyberattacks against the KA-SAT network to Russian Main Intelligence Directorate (GRU)". Viasat's report names no actor (*KA-SAT Attack* §2.2).

The design-basis reading of the three is direct. In each, what decided the consequence was how long the access facilities could keep themselves running, how much of the area's service survived, and how far a single act could reach. Those are the class's measures. At Nashville it was the batteries against the time the operator was kept out; after Maria, the fuel that could be delivered and defended; at KA-SAT, the partition one act could enter and the time to restore a fleet of terminals one by one.

Four reference events of the emergency services series reach this class from the other side of its boundary, and their instruments return findings to it, taken in at v0.2. After the derecho of June 2012 two central offices in Northern Virginia lost power because their generators failed, one not load-tested for at least a year, and the carrier's monitoring of its network lasted thirty minutes after the power failed (*Derecho 911 Outages* F-1, F-2). In October 2016 handsets placed 911 calls when a web page told them to, with no act by their users (*Call Flood* F-1). At Lahaina in August 2023 alerts were sent over a cellular network that was failing, and the people who sent them could not see that they were not arriving (*Lahaina Wildfire* F-1). And in December 2023 an attack on Ukraine's largest carrier took an alert system out of service where its triggering depended on that carrier, while the State Emergency Service's cell broadcast had added a channel that works with minimal signal (*Ukraine Alerting* F-3, F-7).

3. The Facility-Class Design Envelope

3.1 Why Access Is the Second Class

Access is developed second because it is where the elevation rule of the framework fires most often (framework §6.1, §6.3), and because its power dependency is the sector's most immediate coupling with the electric sector (framework §6.2). Transport decides whether a region can reach the rest of the world; access decides whether a person or a facility can reach the network at all.

3.2 The Two Governing Measures

Two measures govern the class.

Backup endurance is how long an access facility keeps providing service on its own power after the grid fails, counting batteries, any generator and the fuel on hand. It is counted for each site and set against the duration of the grid outage the area must be designed for. Endurance that depends on fuel deliveries counts only the fuel on site unless the deliveries are themselves protected and assured. On the author's decision of 7 October 2026, backup power is credited toward backup endurance, for any site counted toward surviving coverage or serving a critical dependent, only when it has been shown to start automatically and to carry its full load for a

stated period, at an interval the operator sets; a generator not so shown counts as none (*Derecho 911 Outages* F-2). Endurance is counted against the longer of the area's design grid-outage duration and the time the operator may be kept from the facility after an event that damages it; at Nashville the batteries lasted about four to five hours and the operator was kept out about twelve (*Nashville Bombing* F-2). A generator not protected against an event counts as no generator for that event, as one not shown to run counts as none, so that a central office's endurance against its own loss is its battery endurance unless its generators are so protected (*Nashville Bombing* F-3). Resupply counts as protected only if it is protected against theft at the site and in transit, and as assured only if the operator's claim on fuel survives the government's allocation of fuel in a declared emergency, by arrangement made before the event; after Maria fuel and generators were stolen and, on one executive's account, federal agencies took control of the fuel supplies, so that local operators could not get fuel when they needed it (*Hurricane Maria* F-3). A broadcast transmitter counted toward surviving coverage for alerts has its endurance counted as a site's, including the analog or simpler path that keeps it on the air (*Hurricane Maria* F-8).

Surviving coverage is the share of an area's access facilities, or of the population and critical dependents they serve, that remains in service at a given time after an event. It is the figure the Federal Communications Commission's Disaster Information Reporting System already reports, county by county, during every activation, and the class adopts it so that the design basis is stated in the sector's own instrumentation. It is stated with its base, the sites and providers counted, and with the population covered and the population able to reach 911 from any network beside the share of sites, for wireline and cable as well as wireless, and it is reported over a channel that survives the event (Section 7.3, CDB-4). After Maria the system's base moved from 1,789 sites to 2,671 between 23 and 27 September as companies were added, each report called itself incomplete, and the figures could be neither reported nor verified while the networks were down (*Hurricane Maria* F-4). Surviving coverage for 911 is the population able to reach any network from which a 911 call is carried (*Hurricane Maria* F-7).

The design basis requires that, for the grid-outage duration an area must withstand, enough access facilities keep running on their own power that surviving coverage never falls below the level needed for people to reach 911 and to receive emergency alerts, and that every critical dependent keeps at least one working access path.

3.3 The Three Consequence Chains

Chain P — power. A wide-area grid outage drains every access site in the area together. Coverage holds while batteries last, then falls as sites go dark, fastest where sites have no generator and where fuel cannot reach. Its size is set by backup endurance against the outage duration, and it is read in surviving coverage. Hurricane Maria realized it across a territory, with the storm and the grid together: on the Government Accountability Office's analysis about half the sites out in September were out for want of power and about four-fifths by October, and repaired sites stayed dark (*Hurricane Maria* F-1).

Chain D — dependent. The loss of one access facility severs a critical dependent whose service passes only through it. The rest of the network may be untouched. Its size is the consequence at the dependent: a 911 center that cannot receive calls, a hospital that cannot reach its systems, a utility that loses sight of its field stations. Nashville realized it, the building's endurance set against the time its operator could not reach it, and its reach counted beyond the building's own dependents (*Nashville Bombing* F-4); after Maria it was realized across a territory where 911 centers answered without the location and call-back information the access network delivers (*Hurricane Maria* F-7).

Chain M — management. A compromise or failure of the systems that manage sites or terminals remotely takes many of them out of service at once. Its size is the number of sites or terminals a single system or a single change can reach, which is the access equivalent of fleet reach in the Chemical series' coordination class (*Chemical Enterprise Coordination and Control* §3.1). The reach of a single act is the management partition it can enter (*KA-SAT Attack* F-1).

3.4 Design Parameters at Class Resolution

The two measures are the class parameters, together with a third: the critical-dependent map, the record of which critical dependents each access facility serves and whether it is their only path. A carrier that cannot produce this map cannot show where its elevation obligations fall. The values depend on the area and are not given here as numbers (Section 10.1).

The critical-dependent map records the reach of each facility: the dependents whose only path passes through it by any route, including the backhaul of other sites (Section 4.4), other carriers' traffic, other providers' lines and other sectors' circuits, counted across the area its loss would reach and not only the area it serves. One building at Nashville carried the backhaul of intact cell sites, another carrier's traffic, FirstNet, the State's network and lines agencies obtained from a second vendor, and AT&T counted 1,166 ZIP codes in seven states as affected (*Nashville Bombing* F-4). How much one building may carry is answered from the map. Where the reach extends into other carriers' networks or other sectors, the map is shared with them under Section 8.1, and the counting of paths behind the building is Transport and Backbone's (*Transport and Backbone* §3.2).

For an access network that serves its users through distributors, the map is complete only when the distributors contribute to it, and a dependent can learn which management partition its terminals share; at KA-SAT dependents such as Enercon held their own terminals on a network sold through distributors, and Viasat's report does not mention the turbines (*KA-SAT Attack* F-6). The map is also held by the electric utility, with a restoration priority for access sites and their backhaul, before an event (Section 7.3, CDB-5; *Hurricane Maria* F-6). Two further parameters are added at this revision: the time of exclusion, the time an operator may be kept from a facility an event has damaged (Section 4.3), and the management partition, the group of sites or terminals one management act can reach (Section 4.6).

4. Design Basis Events: Non-Adversarial Envelope

The framework lists candidate bounding events for the sector, including regional access denial, but does not develop them (framework §4.1). This section develops the events for access under the series' four headings: the event, the harm that makes it count, what must be done, and for how long. Section 5 applies the military-action extension to them.

4.1 Grid Outage Outlasting Backup Power

Postulated event a wide-area loss of grid power lasting longer than the batteries at access sites, with fuel resupply delayed by road closures, fuel shortage or competing demand. A storm that strikes the access network and the grid together is one such event (Section 4.2).

Consequence threshold surviving coverage in an area falling below the level needed to reach 911 and receive alerts, or a critical dependent losing its last access path.

Required function every access facility must state its backup endurance. Sites that serve critical dependents, or that are needed to hold the area's surviving coverage, must carry endurance at least equal to the area's design outage duration, by generator, fixed fuel and protected resupply. The operator must hold a fueling plan that does not depend on the communications the outage has taken away. The endurance stated is the endurance shown by test under Section 3.2. In the derecho one office's generators had not been load-tested for at least a year, one would not start and the other could not carry the load; another office's generator failed in its automatic start (*Derecho 911 Outages* F-2). The fueling plan must not depend on the roads and ports the same event closes; after Maria ports and airports were closed and then constrained, roads were impassable and fuel was scarce, and resupply is credited only as Section 3.2 allows (*Hurricane Maria* F-3).

Design-basis duration the area's design grid-outage duration, which for a major hurricane or a deliberate attack on the grid is days to weeks where the grid can be repaired, and months where it would have to be rebuilt rather than repaired, as an island's after a major hurricane, or where an attacker has damaged its transmission and distribution. After Maria the majority of the main island's grid was down until 17 November 2017 on the Federal Emergency Management Agency's account, outages continued through May 2018, and the access network took six months to reach 4.3 per cent of sites out (*Hurricane Maria* F-2).

4.2 Storm, Wildfire and Flood Damage to Sites

Postulated event wind, wildfire, flood or ice damage to towers, antennas, cabinets, generators and the local lines that connect sites and customers.

Consequence threshold loss of enough sites in one area that surviving coverage falls below its required level.

Required function sites needed for surviving coverage must be hardened to the area's design event, and the operator must hold deployable replacement capacity, such as portable cell sites

and generators, sized to the area. Sites in wildfire areas must keep their equipment and power within a defensible space. A storm that strikes the access network and the grid together is one design-basis event, whose duration at each site is the later of the repair of the site and the restoration of its power, or of a generator and its fuel; after Maria one carrier had repaired 93 per cent of its sites within 100 days and had half of them in service (*Hurricane Maria* F-1). Deployable replacement capacity counts only where it is held in the area or can reach it while its ports and roads are closed (*Hurricane Maria* F-1). Satellite-linked deployables, which share no building, backhaul or power supply with the facility they stand in for, are counted toward deployable capacity at their demonstrated time on air; at Nashville AT&T self-deployed one, on the air within about four hours of the batteries' exhaustion where its contract requires fourteen hours from a request, and had seven up within twenty-four hours of the blast (*Nashville Bombing* F-7).

Design-basis duration the later, at each site, of the time to repair or replace it and the time to restore its power, which after a major storm is weeks and, where the grid must be rebuilt, months (Section 4.1).

4.3 Loss of a Central Office

Postulated event fire, flood, explosion or structural failure at a central office, or loss of its power plant, which may be lost with the building standing (*Nashville Bombing* F-1).

Consequence threshold loss of service to every line and every critical dependent served through the building.

Required function no critical dependent may have all its access paths through one central office. The building's power plant, meaning its batteries, generators and fuel, must be protected against the same event that damages the building, and must not share a location with the point of greatest exposure. This is the lesson of Nashville, where the building stood and its power plant did not (*Nashville Bombing* F-1). The power plant, including the commercial connection, is protected against the fire, flooding, fire suppression and loss of the commercial feed that the same event causes, among them the flooding of the building's own water mains and the water used to fight the fire, as well as against the event itself; and the commercial connection and the generators are placed and connected so that one event at one face of the building cannot disable both (*Nashville Bombing* F-1). They are physically protected, with standoff from the public way: at Nashville a vehicle in the public street reached the power connection and generators without entering the building. The concealment of what a building carries, which AT&T said it would pursue in the building's new façade, is recorded as practice (*Nashville Bombing* F-8). For a 911 center the access paths include the links that deliver the caller's location as well as the voice path, and the diversity recorded for them is verified against the physical route of each circuit at a set interval and on any change; in the derecho one county's three location links all ran through one office, and another's circuits had single points of failure a diversity audit might have found. The verification is shared with Core Switching, Routing and Signaling, which owns the routing

inside the building (*Derecho 911 Outages* F-3). The evidence a carrier supplies to a 911 authority under Section 8.1 covers every line a center relies on to receive calls and to be reached, among them 911 trunks, location links, administrative numbers, alarm-company lines and lines obtained from other providers that ride the carrier's buildings, and shows by test, at the interval above, that the center's calls reach it, with the caller's location, when any one central office is lost. At Nashville lines agencies believed independent, from a primary and a secondary vendor, traversed a single choke point, on the State's after-action supplemental report as the Special Bombing Review Commission states it; the routing of calls to another office is Core Switching, Routing and Signaling's (*Nashville Bombing* F-5). After Maria Puerto Rico's two 911 centers answered throughout without full location and call-back information, the loss of location delivery in Chain D across a territory (*Hurricane Maria* F-7).

Design-basis duration the time to restore service from the building or to move it elsewhere, which may be days to months, and, at the outset, the time of exclusion, the time the operator may be kept from a building an event has damaged. At Nashville AT&T's personnel were kept out for about twelve hours, within perimeters set by the Federal Bureau of Investigation and other law enforcement agencies, and had limited access for about two days (*Nashville Bombing* F-2). The exclusion was an act of public authority, and this class makes no finding on it.

4.4 Loss of Backhaul

Postulated event loss of the fiber or microwave link between an access facility and the network behind it.

Consequence threshold a site that is powered and intact but cannot carry calls.

Required function sites needed for surviving coverage must have backhaul that does not share a single point of failure with other such sites in the area. The paths behind them are counted under *Transport and Backbone* §3.2. A cell site or local network served by one transport path carries satellite backhaul ready to take its traffic before the path is lost, so that the place it serves keeps its surviving coverage while the path is down. The paths behind a place are counted for each populated place to the places it depends on, its own capital among them, and island by island where the places are islands, not only for the region to the rest of the network, as *Tonga Isolation* R-6 asks *Transport and Backbone* to state at its §3.2, which it does at v0.3 (*Tonga Isolation* F-6). When the eruption of January 2022 broke the one domestic cable joining Tonga's capital to its northern island groups (DBA-COMM-RE-3), those islands were on satellite for eighteen months; their local networks survived on satellite backhaul the operators assembled after the event, and the government said in July 2023 that satellite capacity could not meet public demand (*Tonga Isolation* §3.4, F-6). Backhaul that shares the electric grid's poles, or the route the grid's repair will clear, is not independent of Chain P and is counted with the grid as one cause; satellite backhaul ready before the event is the path credited against it. After Maria the fiber that connected sites hung on the utility's poles, of which the National Hurricane Center gives 80 per cent as knocked down (*Hurricane Maria* F-5). Satellite backhaul is itself managed in common

with every terminal of its management partition, and is independent of a terrestrial path only where its management is (Section 4.6; *KA-SAT Attack* F-7). A site whose backhaul passes through another facility is within that facility's reach in the critical-dependent map (Section 3.4).

Design-basis duration the repair time of the backhaul path.

4.5 Radio Interference

Postulated event interference with the radio access network, whether from faulty equipment, unlicensed transmitters or loss of the timing that coordinates base stations. Among the events at the class's edge is interference with mobile network frequencies from across a border. DBA-COMM-RE-12 (*Baltic Navigation Interference*) records at its Section 3 that, during the satellite-navigation interference in the Baltic region from 2023, Finland and Lithuania reported interference to their mobile network frequencies near the border to the Radio Regulations Board (*Baltic Navigation Interference* §3). The class notes the event here and develops no requirement from it.

Consequence threshold loss of usable service across an area while the sites themselves are working.

Required function the operator must be able to detect, locate and report interference, and base stations must hold their timing on their own for a stated period if the timing signal is lost (Section 8.2).

Design-basis duration the time to locate and stop the source, or the timing holdover period.

4.6 Cyber Attack on Site and Terminal Management

Postulated event compromise of the systems that remotely configure, update and power-manage cell sites, central-office equipment or user terminals, or of the vendor channels that update them.

Consequence threshold a single compromise or change that takes out of service more sites or terminals than an area's surviving coverage can lose, or that reaches a critical dependent's only access path.

Required function a management system counts as a shared element, so sites it controls together are not independent against this event. A change that can put many sites or terminals out of service, including a software update or a remote shutdown, must not be distributable to all of them in one operation without a governed act, meaning an authorized and recorded change made for each group of sites the change could take out together (*Local Sovereignty* §3.2). A site's power transfer and its ability to keep serving must not depend on the management system.

The reach of a management act is the size of the management partition it can enter, and the group of sites or terminals for which a governed act is made is bounded by that partition. A partition is sized against what an area's surviving coverage and the critical dependents it serves can lose, so that a critical dependent's terminals, and terminals counted toward surviving coverage, do not share a partition with a mass consumer population. At KA-SAT one set of commands from one

distribution system put tens of thousands of terminals out of service across a continent in about forty-five minutes, and the incident was confined to one consumer-oriented partition, Viasat's directly managed mobility and government users not being affected (*KA-SAT Attack* F-1).

The remote-access path into a management system is part of that system. It admits no authentication by credentials alone, records and alarms unusual attempts, and is separable from the management functions it reaches, so that entering it does not by that act reach the distribution of software to sites or terminals. At KA-SAT the attacker entered the trusted management segment through a VPN appliance, by a misconfiguration on Viasat's report of 2022 and with compromised credentials on its officer's account of 2023, and reached the system that distributes software to terminals within hours (*KA-SAT Attack* F-2).

The governed act is enforced structurally at the distribution system, so that the system cannot send a change to more than one partition's worth of sites or terminals, or an executable to terminals at all, without an authorization that the remote-access credentials do not carry, and the record of the governed act is held where the holder of those credentials cannot alter it. The enforcement applies to a management act whether it comes from a proposing system or from a person at a console. A procedure that an attacker holding the operators' access can follow is not a governed act: the commands that wiped the KA-SAT modems were, in Viasat's word, legitimate (*KA-SAT Attack* F-3).

A network serving terminals of consequence can admit returning terminals against a flood from terminals it does not trust, including against the abuse of the address-assignment protocol by which terminals rejoin; at KA-SAT the flood from modems in Ukraine was renewed in new forms for several weeks, on the operator's account of 2023 (*KA-SAT Attack* F-5). A terminal counted toward surviving coverage or serving a critical dependent can be restored at the terminal to a known image that the management system cannot overwrite, and the operator, or the dependent, holds replacement terminals sized to the partition (*KA-SAT Attack* F-4).

Design-basis duration indefinite; the condition lasts until the compromise is found and removed, and for disabled terminals until they are replaced or restored. The time to restore a fleet of terminals is stated in the terms that set it: the number of terminals reached, the share that can be restored remotely, the stock of replacements and the reach of the people who install them. At KA-SAT the network was stabilized within days, Viasat had shipped nearly 30,000 modems by 30 March 2022, and Enercon restored remote monitoring to over 95 per cent of its turbines by 19 April 2022, about eight weeks after the attack, one measured point (*KA-SAT Attack* F-4).

4.7 Failure of Alert Dissemination

Postulated event a public alert, originated and authorized by an alerting authority, that does not reach the people in the area it names because the broadcast stations or carrier networks that disseminate it have failed, are failing, or carry it over a single path, while the authority cannot see that it is not arriving.

Consequence threshold a population in the path of a hazard that has not received an alert sent for it, where the alerting authority, believing it sent, does not use another channel.

Required function each carrier and broadcast operator that disseminates alerts notifies the alerting authorities of each area it serves of any outage there that affects dissemination, within the warning time of the hazards that area faces, by a means the authority will receive at once and over a path that does not depend on the failed network; this is the carrier side of the delivery state the alerting authority holds as an indication (Emergency Management and Coordination §4.2; *Lahaina Wildfire* F-1, R-2). Cell broadcast, which reaches every phone in an area over the operators' base stations with minimal signal, is counted as a dissemination channel where it is available (*Ukraine Alerting* R-8). No alert trigger — the path by which a siren, a public address system or a relay is set off — runs over a single carrier's network without a second path that shares nothing with it; in December 2023 the alert system of a city went out of service when its carrier was attacked (*Ukraine Alerting* F-3). And the sites that disseminate alerts to an area count toward its surviving coverage for alerts as for calls (Section 3.2). After Maria alerts sent over the wireless network did not reach many people, on Federal Emergency Management Agency officials' account to the Government Accountability Office, and the federal government kept the National Public Warning System's radio stations on generators for over two months; no alerting authority's account of what was sent has been found (*Hurricane Maria* F-8).

Design-basis duration the warning time of the hazards the area faces, which for a wind-driven fire or an incoming missile is minutes.

4.8 Emergency Calls Originated Without a Person

Postulated event a user terminal that places 911 calls without an act by its user, because a web page, an application or malicious software instructs it to, across many terminals at once.

Consequence threshold a flood of calls that consumes the capacity of 911 centers, which may not refuse a call (Public-Safety Answering and Dispatch §4.3).

Required function a user terminal able to originate a 911 call does so only on a human act at the point of origin that presents the number before the call is placed. This is the emergency services sector principle of a human act at the point of origin (Emergency Services Sector Framework §4.2), applied here to the terminal, and the answering class's defense against a generated flood depends on it (*Call Flood* F-1, R-1). It does not restrict emergency calls a person places, by any means the terminal offers, including calls placed by a single deliberate act in an emergency.

Design-basis duration for the life of the terminal's software; a terminal that fails the requirement is corrected by update, which is itself a change placed under Section 4.6.

Table 4-1. Class design-basis event categories.

Event category	Postulated event (class-specific)	Consequence chain	Measure
Grid outage	Outage outlasting batteries; fuel delayed	Chain P	Backup endurance,

Event category	Postulated event (class-specific)	Consequence chain	Measure
			counted against the longer of outage and exclusion; surviving coverage
Storm, wildfire, flood	Damage to towers, cabinets, generators; one event with the grid where both are struck	Chain P	Surviving coverage; later of site repair and power
Central office loss	Fire, flood, explosion, power-plant loss with the building standing	Chain D	Critical-dependent map with the facility's reach; time of exclusion
Backhaul loss	Fiber or microwave link cut; fiber on the grid's poles	Chain P or Chain D	Surviving coverage
Radio interference	Faulty equipment, unlicensed transmitters, timing loss; interference from across a border noted	Chain P	Surviving coverage
Cyber attack on management	Remote shutdown or update across a partition; attack on restoration	Chain M	Sites or terminals in one management partition; fleet restoration time
Alert dissemination failure	Alerts not delivered; outage unseen by the authority; single-carrier trigger	Chain D	Surviving coverage for alerts; notification time
Calls originated without a person	Terminals placing 911 calls on instruction	Chain D	Calls originated per act at the terminal

5. Military-Action Extension: DBA-MA for Access

5.1 MA Extension Applicability Determination

Every access facility of consequence performs the facility-level applicability determination the series makes mandatory (framework §7), under the four-filter applicability gate of DBA-MA-AF (*MA Applicability Framework*) §2 and §3. Any one filter satisfied brings the facility into scope. For this class the filters are met most often by geolocation relative to defense-critical loads and

by adversary targeting logic: a central office or a cell site that is the only path for a military installation, a 911 center or a grid control link is a target because of what depends on it. The determination is recorded, and any scope reduction carries a recorded basis (Section 5.8). The determination for a terminal or site counts the reach of the management that serves it, so that a terminal in an area at peace that shares a management partition with terminals in an area at war is within scope. The KA-SAT attack was aimed, on the governments' account, at Ukraine and its armed forces' command and control, and reached consumers in at least six other countries and turbines in central Europe; its effect on Ukraine's armed forces is contested, and the class makes no finding on it (*KA-SAT Attack* F-8).

5.2 The Four Characteristics Applied to the Class

Applied to access, the four military-action characteristics take specific form. Intentionality: the adversary chooses the access facilities whose loss severs the most important dependents, or attacks the grid knowing that access will follow within hours. Persistence: the adversary may hold access in the systems that manage sites and terminals until it is useful; it persists before the act by preparing credentials and knowledge of those systems, as the KA-SAT attackers did, whose first successful login came the evening before the attack on the operator's account of 2023, the record not showing access held long; and it persists after the act by contesting the restoration, as at KA-SAT, where the flood from terminals in Ukraine continued for weeks (*KA-SAT Attack* F-5). Precision: the adversary takes down the sites and buildings that serve emergency services, government and the grid, and leaves the rest, so that the public network appears to work while the services that matter do not; but precision in the choice of target does not imply precision in reach, and the consequence of a management act is counted across its whole reach (*KA-SAT Attack* F-8). Denial: the adversary jams the radio access network, attacks the fuel supply and the roads that resupply generators, and times the act to coincide with a grid outage, so that the class's endurance and its repair are denied together. The fuel, the generators and the delivery of fuel are the target that finishes a grid denial (*Hurricane Maria* F-3). A strike on a critical access facility is a strike on its power plant, from outside it, and the denial of repair includes the operator's exclusion from the facility, which an actor can prolong by a second device, a threat or an attack on responders (*Nashville Bombing* F-2, F-8). The class validation standard is the state actor who knows which access facilities a region's emergency services depend on.

5.3 Grid Denied, Critical Sites Struck as the Governing MA Scenario

The governing scenario is a compound one. The grid in an area is denied, so every access site begins to run down together, and at the same time the access facilities serving critical dependents are struck, jammed or disabled remotely. It joins the three reference events: Maria's wide-area power loss, Nashville's single building serving many dependents, and KA-SAT's remote reach, combined by an adversary who intends all three. A facility struck in it is struck at its power plant, from outside, with an exclusion an actor can prolong (*Nashville Bombing* F-8); and the grid

denial may be joined to a denial of the sector's reporting, so that operators and the regulator cannot see which sites are running (*Hurricane Maria* F-9).

The scenario sets the class's adequacy test. An area passes if, for the design outage duration, its surviving coverage stays above the level needed to reach 911 and receive alerts, and every critical dependent keeps at least one working access path. An area that cannot pass carries a declared residual (Section 5.7). A facility struck in the way Section 5.2 states is counted as lost for the longer of its battery endurance and the exclusion an actor can impose (*Nashville Bombing* F-8), and the consequence of a remote act is counted across the whole reach of the management it entered (*KA-SAT Attack* F-8).

5.4 The Coordinated Cyber Scenario

The second scenario uses Chain M. The adversary gains access to the management systems of the sites or the terminals and, at a chosen moment, shuts them down, corrupts their software, as at KA-SAT, or disables them permanently; whether any KA-SAT terminals were disabled permanently is contested between Viasat's report and the United Kingdom's release (*KA-SAT Attack* §2.2). The scenario includes the adversary's attack on the restoration, from the terminals themselves: at KA-SAT each mitigation of the flood was followed within days, and later within hours, by a new variant (*KA-SAT Attack* F-5). The class answer is the requirement of Section 4.6: no single act may reach more sites or terminals than the area can lose without a governed act, enforced at the distribution system and bounded by the management partition, and a site's power and service must not depend on the management system. The KA-SAT wind turbines show the property worth having: the equipment the terminals served kept working because it did not need the link to run (Section 8.1).

5.5 Jamming as a Design-Basis Event

This is the first class in the series in which deliberate radio interference is a design-basis event. Jamming denies service without touching a site, can be moved, and can be aimed at the frequencies that emergency services and the public use. The class does not require that jamming be prevented, which is not within an operator's power. It requires that the design basis assume it in the governing scenario, that an area's critical dependents have at least one access path that jamming of the public radio network does not deny, such as a wired line or a separate band, and that operators be able to detect and report jamming promptly.

5.6 The Adequacy Result: Surviving Coverage Condition

The test of whether an area's access protection is adequate is whether it can reach what this document calls the Surviving Coverage Condition. When the grid is lost and outside support cannot arrive, enough access facilities keep running on their own power to hold the area's surviving coverage at the level needed to reach 911 and receive emergency alerts, every critical dependent keeps one working access path, and this holds for the design outage duration and, at a facility an event has damaged, for the longer of that duration and the time its operator may be

kept from it (Section 3.2; *Nashville Bombing* F-2). Surviving coverage for 911 is counted as the population able to reach any network from which a 911 call is carried (*Hurricane Maria* F-7).

The protective functions on which this condition rests at each site are the automatic transfer of the site's load to batteries and the automatic start of its generator when grid power fails. They are to be locally sovereign as FD-LS defines the property (*Local Sovereignty* §3.2). The decision is a measurement against a setpoint; the function is colocated with the site it protects; no system an attacker could reach can stop, delay or trigger its action; and no such system can alter its setpoints, logic or bypass state except through a governed act. *Local Sovereignty* governs and is not restated here. A site whose power transfer or generator start can be disabled from a management system an attacker can reach fails that test. Local sovereignty describes what protects the site when its management systems and the authorization architecture around them have failed. It does not decide whether any action needs human authorization (*Local Sovereignty* §4).

Each such site must also hold its function for a set period with no outside power, communications or support. That requirement is written here in this document's own words. Other instruments take a similar requirement from the DBA-MA parent framework's "local sovereignty principle", but whether that principle is the same thing FD-LS defines is still an open question, so this document does not borrow it (Section 10.4).

5.7 Adequacy and the Declared Residual

In the non-adversarial domain, the Surviving Coverage Condition is achievable with the Section 4 requirements: endurance matched to the design outage at the sites that matter, no critical dependent served by a single building, and power and service that do not depend on remote management. Against a deliberate attacker, adequacy has limits.

Three residuals are declared, in the output form of DBA-MA-DiD (*Defense in Depth*) §3.5. First, a grid outage that outlasts the fuel an area can hold and protect will, in time, darken access sites however well they are built. Beyond that point the residual is the loss of coverage, and the response is emergency communications that do not depend on the public network, held by emergency services and government. Second, sustained jamming of the public radio network across an area cannot be prevented by its operators. The residual is the loss of mobile service in the jammed area, and the response is the wired and separate-band paths of Section 5.5. Third, the time to restore a fleet of terminals disabled through their management, while the management links are under attack from the terminals they manage, cannot be shortened below what the share restorable remotely, the stock of replacements and the reach of the people who install them allow; at KA-SAT some modems were restored by over-the-air update and the rest reset or replaced, and over 95 per cent of one dependent's fleet took about eight weeks, and the response is restoration at the terminal to a protected image and a replacement stock (Section 4.6; *KA-SAT Attack* F-4, F-5). The first residual is reached sooner where resupply is not protected and assured as Section 3.2 requires, because the fuel, the generators and the delivery of fuel are what an

attacker takes to finish a grid denial (*Hurricane Maria* F-3). The response to it, emergency communications that do not depend on the public network, is held in quantity before the event, not sent after it; after Maria the territory's regulator held twenty-two satellite telephones and the Federal Emergency Management Agency's demand exceeded its pre-staged supplies (*Hurricane Maria* F-8, §6.11). These residuals bound the class adequacy claim.

5.8 Consequence-Basis Conditionality Check

Any relaxation of a conservative class parameter — a shorter design outage accepted, a site without a generator credited toward surviving coverage, a critical dependent left with one path — must be justified against the class baseline, and the larger the relaxation the stronger the analytical basis required. A relaxation without a recorded basis is not consistent with the class design basis. The framework adopts the principle for the sector at its §4.1 (Section 10.1).

6. The Dual Design Basis, Stated in Full

6.1 The Two Design Bases

An access facility carries two design bases at once: a physical design basis, defined in this instrument, and an AI-governance design basis, governed by the Foundational Definitions and the Command Broker family and named and referenced here (framework §5). They interact at the point developed in Section 6.4.

6.2 The Physical Design Basis: The Determinism Posture

The physical design basis is the set of protective functions a site relies on to keep serving independently of the management systems. These are the transfer to battery and the start of the generator, the protection of the power plant, the holdover of timing, and the integrity of the alarms and measurements operators see. Power transfer is the clearest locally sovereign function in the class when it is built so: the decision is a loss of voltage measured at the site, and the switch acts there.

Whether a function is deterministic decides whether the authorization rules apply to it at all. It does not decide where its actions fall on the Bright Line. Power transfer and generator control that behave predictably within a stated range of inputs, and that have been formally tested as installed, are exempt from the Command Broker requirement for the actions they take while their inputs stay in that range (*Layer Decomposition and Determinism* §4.3). They are not below the line: they decide whether a site keeps serving, and they are outside the obligation because their behavior is bounded by proof (*The Bright Line* §4.5). A component that has not been shown to meet this test is covered by the rules, and each of its actions is judged on its consequence.

6.3 The AI-Governance Design Basis: Named and Referenced

Machine-learning and generative-AI systems in access include energy-saving systems that switch cells, carriers or radios off when traffic is low; self-organizing network systems that retune cell

settings across an area; systems that schedule and push software to sites and terminals; and operator decision support in the network operations center. They are governed by the Bright Line and Command Broker architecture (framework §5) and by the Command Broker family, whose parent is CB-Framework_Cross_Sector (*Command Broker Framework*). CB-REG (*Command Broker Family Register*) is authoritative for the family's version state. The framework places the implementation of that architecture with the operations-and-coordination class, OpsCo (framework §3.6). This document names the actions in access that the architecture applies to. Placement is by the consequence of each action, not by the component (*The Bright Line* §4.2, §4.3).

Three cases illustrate the application, and none of them is a placement by component. An energy-saving system within scope that switches a cell off takes an action with every switch-off, and each is placed on its consequence. A cell switched off at night may be the only coverage a 911 caller has; a switch-off that could leave any part of an area without coverage for 911 is above the line and requires Command Broker authorization, while a switch-off of one of several overlapping carriers, whose consequence cannot reach that point, is placed below it. A self-organizing network system that retunes many cells at once takes one action whose consequence is the combined change in coverage, and a retune that could drop surviving coverage below its required level, or take a critical dependent's only path, is above the line. A software update or remote command pushed to many sites or terminals is one action placed on what it could do to all of them together, which is the KA-SAT lesson, and a push that could disable more than the area can lose is above the line.

Consistent with CDB-6 of the *Military Action Communications Design Basis*, the loss or degradation of the operator's own communications does not suspend the Command Broker obligation. CDB-6 states the obligation for actions recommended by generative AI; this instrument applies it to every component within the scope of *Layer Decomposition and Determinism* §4.3, which is the scope the Foundational Definitions set.

6.4 The Interaction Point: Coverage Under Denial

The two design bases meet when an area must keep its surviving coverage while under attack. The control system must be architecturally incapable of executing an above-line action originating from a component within the scope of *Layer Decomposition and Determinism* §4.3 without Command Broker authorization. That includes any command that switches off coverage a critical dependent or a 911 caller relies on, retunes an area below its surviving coverage, or pushes a change to many sites or terminals at once. And each site's power transfer and service must keep working even when no outside system or authority can be reached at all. The physical design basis guarantees that sites keep themselves running; the AI-governance design basis guarantees that no proposing system can switch them off without a qualified human authorization the architecture can enforce. Beside that requirement, the distribution system enforces the governed act of Section 4.6 structurally, for a management act from a proposing system or from a person at a console alike (*KA-SAT Attack* F-3).

6.5 Scope Boundary

The physical design basis is defined in full in this instrument. The AI-governance design basis is named and referenced, not redefined; its governing text is *The Bright Line* with its determinations, and the Command Broker family under its register. Where the two are cited together, the physical requirement governs the access protective architecture and the AI-governance requirement governs the authorization of any proposing system that interacts with it.

7. The Communications Design Basis Applied to Access

7.1 Why It Applies

An access facility delivers other sectors' communications and also depends on communications of its own. Its management systems watch and configure its equipment, its alarms report its state, including how much battery it has left, and its crews coordinate fueling and repair, all over communications links. The *Military Action Communications Design Basis* applies to those links as it applies to any facility (*Military Action Communications Design Basis* §3). This section states how each of the seven requirements applies to access. It cites them; their text is in that document.

7.2 The Finding That Shapes the Rest: A Dark Site Cannot Report

A site's alarms and management traffic travel on the same backhaul as its service. When the backhaul is cut, or the site's power fails, the operator loses sight of the site at the moment it most needs to know whether the site is running, how long its batteries will last and whether its generator has started. Fueling crews are then dispatched on guesses. A management link that shares the site's backhaul is not an independent channel. The requirements below follow from this finding. The finding reaches the sector as a whole. After Maria providers could not report to the Disaster Information Reporting System because they had lost their communications, carriers learned that an area had power by going there, and the Federal Emergency Management Agency could not see the state of 37 of 69 hospitals a week after landfall (*Hurricane Maria* F-9). At Nashville, while the building ran on batteries its operator knew would fail, no known notice reached the 911 centers, on the State's after-action supplemental report as the Special Bombing Review Commission states it (*Nashville Bombing* F-6).

7.3 The Seven Requirements

CDB-1, degradation as baseline. The access design basis assumes from the outset that the operator's own management, alarm and coordination links are degraded or lost in any event that causes a wide-area outage. Fueling, restoration and deployment plans are written and tested against that condition. The degraded baseline includes management links under attack from the terminals they manage, as at KA-SAT (*KA-SAT Attack* F-5), and the loss of the sector's own reporting to its regulator, as after Maria (*Hurricane Maria* F-9).

CDB-2, local protective and safety autonomy. This requirement is still worded for the electric sector, where it names a grid's supervisory control and energy-management systems. Read for access, it requires that loss of the management link not cause loss of the site's power transfer, generator start or ability to keep serving, and that a site never shut itself down because it has lost contact with its management system. Section 5.6 states the property, and the design basis must demonstrate it rather than assume it. The KA-SAT turbines confirm the reading: they kept running in automatic mode when the link was lost (*KA-SAT Attack* F-7).

CDB-3, field-executable emergency procedures. Crews must be able to fuel, restart, reconfigure and replace sites, and to deploy portable sites, without direction from the network operations center, from procedures and site records held in a form that does not depend on the network. The procedures include a protocol, agreed in advance with law enforcement and the authority having jurisdiction, for the operator's entry or remote action to preserve power at a damaged facility that is a crime scene, as AT&T said it was pursuing with the Federal Bureau of Investigation and the Federal Communications Commission after Nashville (*Nashville Bombing* F-2); the restoring and replacing of terminals by the people who hold them, among them distributors and dependents (*KA-SAT Attack* F-4); and the assessment of site damage and power state without the network (*Hurricane Maria* F-9).

CDB-4, verified independent channels. Each central office serving a critical dependent, and each site needed for an area's surviving coverage, must report its power state over a channel that does not share its backhaul. Satellite and separate-band radio are the usual choices. The detection of a failure of 911 call completion or of alert dissemination through a facility, and the notification of the 911 centers and alerting authorities it serves, are powered and connected independently of that facility for the area's design outage duration, and notification reaches them at once, not by e-mail to designated staff; in the derecho the carrier's monitoring of its Northern Virginia network lasted thirty minutes after the power failed, and it learned from a county that the county's 911 service was down (*Derecho 911 Outages* F-1). The management channel is reachable only by a route the attacker cannot borrow, not by the remote access that serves the operators' ordinary work (*KA-SAT Attack* F-2). A facility serving 911 centers or alerting authorities that is running on batteries notifies them, while it is still running, of the expected time of failure; the path by which the carrier notifies a 911 center or an alerting authority, and the path by which the authority reaches the carrier, are each shown not to pass through the facility whose failure is reported, and receipt is confirmed. At Nashville the carrier's notification system worked as designed and did not reach answering points whose links the outage had damaged, and some received nothing for six days, on the Tennessee Emergency Communications Board's finding as reported and the State's after-action supplemental report (*Nashville Bombing* F-6); the pattern of late and incomplete notification by a carrier has its home in DBA-ESS-RE-2 (*Multistate 911 Outage*) F-7. The independent channel also serves the operator's report to the regulator and to the Commission's reporting system, and is held in a number sized to the area before the event (*Hurricane Maria* F-9). A broadcast transmitter counted toward surviving coverage for alerts

reports its state over such a channel; after Maria broadcasters did not report their state, which the Commission learned by scanning the spectrum (*Hurricane Maria* F-8).

CDB-5, coordination without shared connectivity. This requirement is also still worded for the electric sector, where it names the gas–electric boundary. Read for access, it requires that carriers serving one area, the tower companies that host their sites, the electric utility restoring power, and the 911 authorities and the carriers' outage contacts be able to coordinate fueling, access, restoration and notification when their usual channels are down, with contacts kept current at an interval shorter than a year (*Nashville Bombing* F-6). It requires also that the coordination of access with the electric utility rest on a restoration priority for access sites and their backhaul, and for the critical-dependent map of Section 3.4, held by the utility before the event, and on a protection of reinstalled communications plant during the grid's debris clearance, as well as on a channel that survives the event; the grid's side of both is the energy series' to state (Section 8.3). After Maria coordination among carriers and with the regulator ran in person from landfall, the territory's regulator meeting the companies daily for about sixty days, while coordination with the electric utility failed for want of a shared priority, its crews removing newly installed fiber during debris cleanup (*Hurricane Maria* F-6).

CDB-6, Command Broker continuity. Section 6.3 applies it.

CDB-7, verification cadence. The independent channels of CDB-4 are tested at set intervals with the primary links assumed lost, and the test includes a fueling and restoration exercise run without them. The verification includes reporting to the regulator with the primary links lost (*Hurricane Maria* F-9).

8. Cross-Sector Dependencies and Regulatory Context

8.1 What Other Sectors Depend On: Critical-Services Elevation at the Access Boundary

Access is where the framework's elevation rule fires most often (framework §6.1, §6.3), and this class is where the critical-services overlay takes effect. An access facility that is the sole path for a 911 call center, a hospital, a station carrying the Emergency Alert System, the priority services used by government and critical infrastructure, or the control link of a facility carrying a Tier-2 or Tier-3 designation in another DBA series inherits a tightened acceptance criterion. For this class that criterion has two parts: backup endurance at least equal to the area's design outage duration, and a second access path that shares no building, backhaul, power supply or management system with the first (*KA-SAT Attack* F-7). Seen from the dependent facility, the loss of its access is the onset of the degraded baseline that CDB-1 describes, and the KA-SAT wind turbines show a dependent whose protection and grid control were built to the degraded baseline and whose monitoring was not: the turbines ran in automatic mode and the grid operators kept unrestricted access to them, but their remote monitoring and maintenance were lost for about eight weeks for over 95 per cent of them, and Enercon named the backup link that

many wind farms lacked as the problem (*KA-SAT Attack* F-7). Where a carrier serves a 911 authority or an alerting authority, it supplies on the authority's request the evidence of backup power, circuit diversity and monitoring for the facilities that serve it, since the emergency services sector holds its own evidence of the elements it does not own rather than resting on a certification made to a regulator, the annual federal certification having become a one-time filing in June 2026 (Emergency Services Sector Framework §4.2; *Derecho 911 Outages* F-5). The communications framework keeps the critical-services overlay as an elevation overlay with no instrument of its own (Communications Sector Framework §3.7), and this section is where it takes effect.

The elevation fires on the reach of a facility recorded in the critical-dependent map, not only on the dependents it serves directly (Section 3.4; *Nashville Bombing* F-4), and on a dependent's control or monitoring link however the dependent buys its terminal, including as a consumer service (*KA-SAT Attack* F-6). FirstNet, any priority service and any radio or data system a dependent relies on as a backup counts toward its second path only where it shares no building, backhaul, power supply or management system with the first, the test Section 5.5 applies to jamming here applied to the loss of a building, with the management system added (*KA-SAT Attack* F-7). At Nashville FirstNet, whose infrastructure was not damaged, lost service because it shared the building's power, and the State's radio and dispatch systems, which rode the State's network mainly provided by AT&T, did not work (*Nashville Bombing* F-7). For a 911 center the second path carries location and call-back information as well as voice (*Hurricane Maria* F-7), and the evidence a carrier supplies covers every line the center relies on and is shown by test (Section 4.3; *Nashville Bombing* F-5).

8.2 What Access Depends On

Access depends first on electric power and on the fuel and roads that keep its generators running (framework §6.2). It depends on backhaul, which is counted under *Transport and Backbone*. It depends on timing: base stations draw timing from satellite signals to coordinate their transmissions, and lose service when timing fails and their holdover runs out; their timing receivers take their design basis from *Positioning, Navigation and Timing* §7, on the author's decision of 6 October 2026 that the timing receiver design basis for every sector is held there and cited. And it depends on tower companies, which host the equipment of several carriers on one structure, so that one tower can be a shared place for several networks. A building carrying several carriers' traffic is counted, as a shared tower is, as one place for all of them (*Nashville Bombing* F-4).

8.3 Boundaries

With Core. On the author's decision of 6 October 2026, a central office building and its backup power belong to this class, and the switching and routing decisions made inside it belong to DBA-COMM-CR-FC3. The building's protection is therefore stated here, and the consequence of a switching action taken inside it is placed under Core.

With Transport. Paths that carry traffic between an access facility and the backbone belong to DBA-COMM-TR-FC1 from the point where local traffic is gathered onto them, and backhaul is counted under *Transport and Backbone* §3.2. Backhaul on the grid's poles is counted with the grid under Section 4.4, and the counting of shared poles as a shared element at Transport and Backbone §3.2 is that class's (*Hurricane Maria* F-5; Section 10.9).

With timing. The reception, holdover and distribution of timing is DBA-COMM-PN-FC4's. This class requires only that each base station state its own holdover.

With the critical-services overlay. The overlay names the services; this class states the elevation where they reach the network (Section 8.1).

With Emergency Services. The boundary for 911 lies where the call and the caller's location enter the 911 system: the originating networks and, in an enhanced 911 system, the 911 trunks, selective routing and location delivery are this series', and the 911 center and, in a Next-Generation 911 system, the Emergency Services IP Network and its core services are the emergency services series' (Public-Safety Answering and Dispatch §8.3). This class protects the access paths that reach them. For alerts, origination is the alerting authority's and dissemination is this class's (Section 4.7; Emergency Management and Coordination §8.3). After Maria it was this series' delivery of location that failed while the centers answered (*Hurricane Maria* F-7).

With Energy. Grid restoration and the control links of the grid are the energy series'. The power dependency runs both ways (framework §6.2), and the energy series' own instruments decide how they count access as a dependency. The restoration priority the utility holds for access sites, their backhaul and the critical-dependent map, and the protection of reinstalled plant during debris clearance, are the energy series' to state on the grid's side (Section 7.3, CDB-5; *Hurricane Maria* F-6).

8.4 Regulatory and Institutional Context

The Federal Communications Commission collects outage reports through its Network Outage Reporting System and, during disasters, cell-site status through its Disaster Information Reporting System, whose county-level figures this class adopts as its measure of surviving coverage. The Commission sets reliability requirements for providers of 911 service, rules for the Emergency Alert System, and a mandatory disaster response framework, adopted in 2022, under which wireless providers cooperate during emergencies and which binds roaming and mutual aid (*Hurricane Maria* F-4). Some states set backup power requirements for access facilities; California requires seventy-two hours of backup power at wireless sites in its highest fire-threat areas. FirstNet provides a public-safety broadband network with its own priority. Reporting to the Disaster Information Reporting System has since been made mandatory, as revised in May 2026 (*Hurricane Maria* F-4). For the access paths of satellite networks the advisory that bears is AA22-076A of 17 March 2022, which recommended for satellite providers multifactor authentication, least privilege, removal of unnecessary trust relationships and monitoring of unusual login attempts; it sets no engineering requirement (*KA-SAT Attack* F-2).

None of these sets an engineering design basis for backup endurance or surviving coverage across an area under deliberate attack. That is the gap this class fills (framework §2.3).

9. Inheritance Map and Anticipated Derivatives

9.1 The Inheritance Map

Table 9-1. Inheritance map for DBA-COMM-AC-FC2.

Element	Source	Inheritance mode
Function-primary taxonomy and the class's place in it	framework §3.1, §3.3, §8.2	Inherited unchanged
Power dependency as a defining characteristic	framework §3.3, §6.2	Inherited and developed (Sections 2.1, 3.2)
Consequence-severity inclusion criterion and candidate events	framework §4.1	Inherited and developed (Section 4)
Bright Line and Command Broker architecture	framework §5; FD-BL; FD-BR	Applied; not restated (Section 6)
Determinism as scope	FD-LD §4.3	Applied (Section 6.2)
Local sovereignty of power transfer and generator start	FD-LS §3.2	Cited directly (Section 5.6)
Communications design basis CDB-1 to CDB-7	DBA-MA-CDB §2, §3	Cited and applied (Section 7)
MA applicability gate	DBA-MA-AF §2, §3	Applied (Section 5.1)
Declared residual	DBA-MA-DiD §3.5	Cited directly (Section 5.7)
Elevation rule and critical-services overlay	framework §6.1, §6.3, §3.7	Applied (Section 8.1)
Backhaul path counting	*Transport and Backbone* §3.2	Cited (Section 4.4)
Central office boundary with Core	author's decision of 6 October 2026	Stated here (Sections 1.2, 8.3)
Reference events	framework §9.1; determination of 9 October 2026	Conferred; instruments written and held by Project 11 (Sections 2.3, 9.2, 10.5)
Satellite backhaul at a site served by one transport path; paths counted place by place	*Tonga Isolation* F-6; *Transport and Backbone* §3.2	Stated here (Section 4.4)

Element	Source	Inheritance mode
Consequence-basis conditionality	framework §4.1	Stated (Section 5.8); adopted by the framework (Section 10.1)
Timing receiver design basis	*Positioning, Navigation and Timing* §7	Cited (Section 8.2)
Endurance counted against the exclusion; a generator unprotected against an event counts as none; the power plant protected against the event's own effects	*Nashville Bombing* F-1 to F-3, F-8	Stated here (Sections 2.1, 3.2, 4.3, 5.2, 5.3)
The facility's reach in the critical-dependent map; second paths, FirstNet and 911 evidence	*Nashville Bombing* F-4, F-5, F-7; *Hurricane Maria* F-7; *KA-SAT Attack* F-6, F-7	Stated here (Sections 3.4, 4.3, 8.1)
Storm and grid as one event; design outage of months; resupply protected and assured; backhaul on the grid's poles	*Hurricane Maria* F-1 to F-3, F-5	Stated here (Sections 3.2, 4.1, 4.2, 4.4)
Coverage with its base and population; reporting and coordination without the network	*Hurricane Maria* F-4, F-6, F-8, F-9; *Nashville Bombing* F-6	Stated here (Sections 3.2, 7.2, 7.3)
Management partition, the remote-access path, the governed act enforced at the distribution system, fleet restoration and the attack on restoration	*KA-SAT Attack* F-1 to F-5, F-8	Stated here (Sections 4.6, 5.1, 5.2, 5.4, 5.7)
Interference with mobile network frequencies from across a border	*Baltic Navigation Interference* §3	Noted (Section 4.5)
Elements addressed to the framework, other classes and other series	the three reference events	Noted (Section 10.9)
Backup endurance, surviving coverage, critical-dependent map, time of exclusion, management partition, Surviving Coverage Condition	—	Defined here

9.2 Anticipated Derivatives

The class anticipates two derivatives, and both are written. The use case for the governing scenario of Section 5.3 is DBA-COMM-UC-2 (*Grid Denied and Critical Access Sites Struck*). Its figures for surviving coverage are to be taken from the record, with their base and the population

covered, rather than from a headline percentage (*Hurricane Maria* F-4); that, and the use-case material Project 11 offered from the three reference events, are for it at its next revision. The reference-event instruments for the Nashville bombing, Hurricane Maria and the KA-SAT attack are written as DBA-COMM-RE-4, DBA-COMM-RE-5 and DBA-COMM-RE-6, conferred under the framework's §9.1 and held by Project 11 under the determination of 9 October 2026, and record what is established about each event (Section 2.3).

10. Gaps and Open Items

10.1 Quantitative Design Parameters

Backup endurance, surviving coverage and the critical-dependent map are stated as parameters. The author accepted the two measures on 6 October 2026 as a good start rather than a final pair, and they may be joined or refined as the analysis matures. The level of surviving coverage needed to reach 911 and receive alerts is still to be set, and the framework adopts the conditionality principle of Section 5.8 for the sector at its §4.1, with the trigger that a path count of one accepted after the loss of a sole path carries a recorded basis (*Tonga Isolation* F-7). On the author's decision of 7 October 2026 two further values are left open with their rules stated: the interval at which backup power is tested for credit (Section 3.2), and the time within which an outage affecting alert dissemination is notified, which is bounded by the warning time of the hazards an area faces (Section 4.7). This revision adds two values left open with their rules stated: the time of exclusion, against which endurance is counted (Sections 3.2, 4.3), and the size of a management partition, which is set against what an area and its critical dependents can lose (Section 4.6).

10.2 Design-Basis Durations

The design grid-outage duration for an area is the duration that governs the class, and no figure is given here. It should be set together with the energy series' own assumptions about restoration after a deliberate attack on the grid, so that the two series assume the same outage. Hurricane Maria is given to the energy series as a point for the outage the two series design for: the majority of the main island's grid was down from landfall on 20 September until 17 November 2017, and outages continued through May 2018 (Section 4.1; *Hurricane Maria* F-2). The grid's side is that series' to state.

10.3 The Boundary with Core

The central office boundary is confirmed by the author and stated in Sections 1.2 and 8.3. DBA-COMM-CR-FC3 adopts it when it is written, and the framework states it at its §3.3 and §3.4 from v0.7.

10.4 Local Sovereignty in the DBA-MA Sense

The protective functions at Section 5.6 cite *Local Sovereignty* §3.2 directly. The endurance requirement beside them is the property the DBA-MA parent framework's local sovereignty principle describes. Whether that principle is FD-LS's construct at all is held pending Project 5's answer, and the requirement is written here in this document's own words until then. The item is registered under Project 9 OI-16.

10.5 Reference Events

The Nashville bombing of December 2020, Hurricane Maria in September 2017 and the KA-SAT attack of February 2022 are conferred by the framework at its §9.1 as DBA-COMM-RE-4 to DBA-COMM-RE-6. Their instruments are written, and under the determination of 9 October 2026 that gives reference events their own project they are held by Project 11. This revision takes in what they return: three corrections of the class's account of the events, from *Nashville Bombing* F-9, *Hurricane Maria* F-10 and *KA-SAT Attack* F-9, at Sections 2.3, 4.3, 5.2, 5.4 and 8.1, and *Nashville Bombing* R-1 to R-8, *Hurricane Maria* R-1 to R-9 and *KA-SAT Attack* R-1 to R-8. The revision of Section 7.3's statement that coordination after Maria depended on the networks that had failed is *Hurricane Maria* R-6, carried by its F-6. The findings of four emergency services instruments that reach this class were taken in at v0.2 (Section 2.3), and v0.3 took in the access element of *Tonga Isolation* R-6 (Section 4.4), from the transport class's reference event DBA-COMM-RE-3. Interference with mobile network frequencies from across a border, routed to this class from DBA-COMM-RE-12 (*Baltic Navigation Interference*), is noted among the events at its edge (Section 4.5).

10.6 Two Communications Design Basis Requirements Worded for the Electric Sector

CDB-2 and CDB-5 are read for access at Section 7.3 because their text names the electric sector. When the *Military Action Communications Design Basis* words them for every sector, Section 7.3 is revised to cite the new wording.

10.7 Energy Saving and Coverage

Energy-saving systems that switch cells off are spreading quickly and are placed in Section 6.3 on the consequence of each switch-off. Whether carriers can produce, for each switch-off, the coverage it removes, which the placement requires, is not established and is the first thing a review of a carrier's practice should test.

10.8 Findings Taken From the Emergency Services Series

Version 0.2 took in *Derecho 911 Outages* R-7 (monitoring independent of the facility's power, backup power credited only when tested, location-link diversity verified against the route), *Call Flood* F-1 and R-1 (the origin obligation at the terminal), *Lahaina Wildfire* R-2 (notification of alerting authorities within the warning time) and *Ukraine Alerting* R-8 (cell broadcast as a

channel and the single-carrier trigger), together with the emergency services sector principle of a human act at the point of origin. Core Switching, Routing and Signaling takes the routing side of *Derecho 911 Outages* R-7 at its next revision.

10.9 Elements for Other Instruments

The three reference events address elements to other instruments, which are noted here and not done. To the Communications Sector Framework: the naming of the loss of terminals through common management among its candidate bounding events at its §4.1, beside regional access denial (*KA-SAT Attack* R-1), and the restating of the central-office backup figure at its §3.3 as a battery figure and a generator figure (*Nashville Bombing* R-3). To Core Switching, Routing and Signaling: the routing of 911 calls to another office when one central office is lost (*Nashville Bombing* R-5). To Transport and Backbone: the counting of shared poles as a shared element at its §3.2 (*Hurricane Maria* R-5). To Operations and Coordination: the notification of 911 authorities during a carrier outage, with trigger points and timelines agreed with them, as a coordination function prepared in advance (*Nashville Bombing* R-6), and the restoration priority among carriers, utility and regulator as such a function, with the ending of disaster roaming recorded among the commercially conflicted decisions of its §6.5 (*Hurricane Maria* R-6). To the energy series: the dependents reached through distributors and the second path outside common management (*KA-SAT Attack* R-6, R-7), the design outage (*Hurricane Maria* R-2), and the grid's side of the restoration priority and of the protection of reinstalled plant (*Hurricane Maria* R-6). To the emergency services series: the evidence of 911 diversity and the notification of 911 centers (*Nashville Bombing* R-5, R-6), and the consequence at the centers of the loss of location delivery (*Hurricane Maria* R-7).

References

DBA-COMM Series and Parent Framework

Communications Sector Framework (DBA-COMM_Sector_Framework). Eclectic Technologies, 2026. Parent sector framework under which this instrument is issued.

Transport and Backbone (DBA-COMM-TR-FC1). Eclectic Technologies, 2026. Backhaul path counting (§3.2).

The Tonga Isolation, January 2022 (DBA-COMM-RE-3). Eclectic Technologies, 2026. A reference event of the transport class, cited by the short title *Tonga Isolation* and finding number.

The Nashville Bombing, December 2020 (DBA-COMM-RE-4), *Hurricane Maria, September 2017* (DBA-COMM-RE-5) and *The KA-SAT Attack, February 2022* (DBA-COMM-RE-6). Eclectic Technologies, 2026. The class's three reference events, held by Project 11, cited by the short titles *Nashville Bombing*, *Hurricane Maria* and *KA-SAT Attack* and finding, recommendation or section number.

Satellite-Navigation Interference in the Baltic Region, from 2023 (DBA-COMM-RE-12). Eclectic Technologies, 2026. A reference event of the timing class, held by Project 11, cited by the short title *Baltic Navigation Interference* for the facts of its Section 3.

Core Switching, Routing and Signaling (DBA-COMM-CR-FC3), *Operations and Coordination* (DBA-COMM-OpsCo) and *Grid Denied and Critical Access Sites Struck* (DBA-COMM-UC-2). Eclectic Technologies, 2026. The core and coordination classes, to which elements are noted (Section 10.9), and the class's use case (Section 9.2).

DBA-MA Parent Framework. Eclectic Technologies. Source of the four military-action characteristics.

Military Action Communications Design Basis (DBA-MA-CDB). Eclectic Technologies, 2026. CDB-1 through CDB-7 (§2) and the instantiation requirement (§3).

MA Applicability Framework (DBA-MA-AF). Eclectic Technologies, 2026. The four-filter applicability gate (§2, §3).

Defense in Depth (DBA-MA-DiD). Eclectic Technologies, 2026. Declared-residual form (§3.5).

Other Series

Chemical Enterprise Coordination and Control (DBA-CHEM-CorpCo). Eclectic Technologies, 2026. Fleet reach, adapted here as the reach of a common management system.

Positioning, Navigation and Timing (DBA-COMM-PN-FC4). Eclectic Technologies, 2026. Timing receiver design basis for every sector (§7).

Emergency Services Sector Framework (DBA-ESS_Sector_Framework), Public-Safety Answering and Dispatch (DBA-ESS-AD-FC1) and Emergency Management and Coordination (DBA-ESS-EMCo). Eclectic Technologies, 2026. The 911 and alerting boundaries (§8.3 of each class) and the sector principles (framework §4.2).

The 911 Outages After the Derecho of June 2012 (DBA-ESS-RE-1), *The Multistate 911 Outage of April 2014* (DBA-ESS-RE-2), *The 911 Call Flood of October 2016* (DBA-ESS-RE-3), *The Warning Failure in the Lahaina Wildfire* (DBA-ESS-RE-7) and *Air-Raid Alerting in Ukraine, 2022 to 2026* (DBA-ESS-RE-9). Eclectic Technologies, 2026. Cited by the short titles *Derecho 911 Outages*, *Multistate 911 Outage*, *Call Flood*, *Lahaina Wildfire* and *Ukraine Alerting* and finding or recommendation number.

Foundational Definitions and Command Broker Family

FD-BL, FD-BR, FD-LD, FD-LS, FD-EV, FD-BL-D1 and FD-BL-D2, cited by designation; version state in FD-REG (*Governing Stack Register*). CB-Framework_Cross_Sector (*Command Broker Framework*), cited by designation; version state in CB-REG (*Command Broker Family Register*).

Regulatory, Standards and Events

Federal Communications Commission Network Outage Reporting System and Disaster Information Reporting System; Federal Communications Commission rules on 911 reliability, the Emergency Alert System and the mandatory disaster response framework; California Public Utilities Commission backup power requirements for wireless facilities in high fire-threat areas; FirstNet; joint cybersecurity advisory AA22-076A, 17 March 2022.

The sources of the three reference events are those their instruments name, among them, for Nashville, the Federal Bureau of Investigation's review of 15 March 2021 and the Special Bombing Review Commission's final report of 29 March 2022; for Maria, the Federal Communications Commission's report on the 2017 Atlantic hurricane season with its Disaster Information Reporting System status reports, and the Government Accountability Office's report GAO-21-297; and for KA-SAT, Viasat's incident report of 30 March 2022, the attributions of 10 May 2022 and Enercon's statements.

Revision History

Version	Date	Author	Description
0.1 — DRAFT	October 2026	Tim Roxey	Initial issue. The second facility class of the DBA-COMM series, typed physical under the author's determination of 30 September 2026. Defines backup endurance and surviving coverage as the governing measures, accepted by the author on 6 October 2026, with the critical-dependent map as the third parameter; adopts the Disaster Information Reporting System's county figures as the measure of surviving coverage; proposes the Nashville bombing, Hurricane Maria and the KA-SAT attack as reference events; takes grid denial combined with strikes on critical sites as the governing military-action scenario and makes jamming a design-basis event; places energy-saving cell switch-offs on their consequence; states the critical-services elevation at the access boundary; cites the communications design basis and applies its seven requirements to access's own management links; states the central office boundary with Core on the author's decision of 6 October 2026. Registers the DBA-MA-sense endurance requirement under Project 9 OI-16.
0.2 — DRAFT	October 2026	Tim Roxey	Takes in the findings four emergency services reference events return to this class, on the author's decisions of 7 October 2026: mobile handsets in scope for their emergency-call and alert-receiving functions, and alert dissemination stated as in the class (§1.2); backup power credited only when shown to run, for any site counted toward coverage or serving a critical dependent (§3.2, §4.1; *Derecho 911 Outages* F-2); location-link diversity verified against the physical route (§4.3); a new design-basis event for the failure of alert dissemination, with notification of alerting authorities within the warning time, cell broadcast as a channel and

Version	Date	Author	Description
			no single-carrier alert trigger (§4.7; *Lahaina Wildfire* R-2, *Ukraine Alerting* R-8); a new design-basis event for emergency calls originated without a person, applying the emergency services sector principle of a human act at the point of origin (§4.8; *Call Flood* R-1); detection and notification independent of the facility's power (§7.3); evidence supplied to 911 and alerting authorities (§8.1). Reference events stated as conferred (DBA-COMM-RE-4 to RE-6); timing receivers cite Positioning, Navigation and Timing §7; the 911 and alerting boundaries restated from the emergency services series. Issued under framework v0.12. Values of the new intervals left open (§10.1). Revision-history entries are left as records.
0.3 — DRAFT	October 2026	Tim Roxey	Takes in the access element of *Tonga Isolation* R-6 (DBA-COMM-RE-3): a cell site or local network served by one transport path carries satellite backhaul ready to take its traffic, and the paths behind a place are counted for each populated place to the places it depends on, island by island (§4.4). States the class's three reference events, the Nashville bombing, Hurricane Maria and the KA-SAT attack, as conferred by the framework at its §9.1 as DBA-COMM-RE-4 to RE-6, their instruments not yet written (§2.3, §9.2, §10.5). Records that the framework adopts the conditionality principle at its §4.1 (§5.8, §10.1). Timing receivers cite *Positioning, Navigation and Timing* §7 by short title (§8.2). The intake of the emergency services findings is dated to v0.2 (§2.3, §10.5, §10.8). §10.3 records that the framework states the boundary with Core at its §3.3 and §3.4 from v0.7. Inheritance map and References updated. An independent check of the new text against the instruments raised 4 items, 4 taken before issue. Conformed to the house style of 7 October 2026. Issued under framework v0.14. Revision-history entries are left as records.
0.4 — DRAFT	October 2026	Tim Roxey	Takes in what the class's three reference-event instruments return, now written and held by Project 11: three corrections of the class's account of the events, from *Nashville Bombing* F-9, *Hurricane Maria* F-10 and *KA-SAT Attack* F-9, at Sections 2.3, 4.3, 5.2, 5.4 and 8.1, with the KA-SAT attribution stated as the European Union's and the named governments', each in its own words; and *Nashville Bombing* R-1 to R-8, *Hurricane Maria* R-1 to R-9 and *KA-SAT Attack* R-1 to R-8. The three events are stated as conferred at the framework's §9.1, written, and held by Project 11 under the determination of 9 October 2026 (§2.3, Table 9-1, §9.2, §10.5). A central office's backup is stated as a battery figure and a generator figure (§2.1). Backup endurance is counted against the longer of the design outage and the time the operator may be kept from a damaged facility, a generator unprotected against an event counts as none, resupply is credited only when protected against theft and

Version	Date	Author	Description
			assured against emergency allocation, and a broadcast transmitter's endurance is counted as a site's; surviving coverage is stated with its base and population, and for 911 as the population able to reach any network (§3.2, §5.6). The critical-dependent map counts a facility's reach, is completed through distributors and is held by the utility (§3.4). A storm striking the network and the grid is one event; the design outage is months where the grid must be rebuilt; deployables count only where they can reach the area, and satellite-linked ones at their time on air (§4.1, §4.2). The power plant is protected against the event's own effects and placed so one event at one face cannot disable it, with standoff from the public way; 911 evidence covers every line by test; the time of exclusion is a duration (§4.3). Backhaul on the grid's poles is counted with the grid and satellite backhaul with its partition (§4.4). The management partition bounds the governed act, which is enforced at the distribution system; the remote-access path is part of the management system; returning terminals are admitted against a flood; fleet restoration is stated in its terms (§4.6). The extension counts the reach of management in applicability, persistence before and after the act, the fuel as the target that finishes a grid denial, the strike on the power plant and the prolonged exclusion, the attack on restoration and a third declared residual, for fleet restoration (§5.1 to §5.7). Section 7.3 reads CDB-1, CDB-3, CDB-4, CDB-5 and CDB-7 accordingly, CDB-5's statement about Maria revised under *Hurricane Maria* R-6. The second path shares no management system, FirstNet and backup radio count only where they share nothing with the first, and elevation fires on reach and on terminals bought as a consumer service (§8.1); AA22-076A and mandatory reporting are recorded (§8.4). Interference with mobile network frequencies from across a border is noted at the class's edge, from *Baltic Navigation Interference* §3 (§4.5). Elements for the framework, Core Switching, Routing and Signaling, Transport and Backbone, Operations and Coordination and the energy and emergency services series are noted (§10.9); use-case material is left for DBA-COMM-UC-2 at its next revision (§9.2). Two values are left open with their rules (§10.1). Tables 4-1 and 9-1 and the References updated. An independent check of the revision against the three instruments, their errata and the routing of batch 10 raised 10 items for this class: 9 taken before issue, and one, the spelling "per cent", which the class and its predecessors use, referred to the author. Issued under framework v0.14. Revision-history entries are left as records.